

Dissecting Conglomerate Valuations

OLIVER BOGUTH, RAN DUCHIN, and MIKHAIL SIMUTIN

ABSTRACT

We develop a new method to estimate Tobin's Qs of conglomerate divisions without relying on standalone firms. Divisional Qs differ considerably from those of standalone firms across industries, over time, and in their sensitivity to economic shocks. The differences are explained by intraconglomerate covariance structures and access to internal capital markets that mitigate external financing frictions. Consequently, the Qs capture variation in the allocation of assets in the economy: within firms through internal capital markets and across focused and diversified firms through diversifying acquisitions. Overall, our method provides opportunities to study the economic mechanisms that explain corporate diversification.

MOST LARGE U.S. FIRMS OPERATE in multiple divisions. Such conglomerates account for the lion's share of financing, investment, and value of U.S. companies. Consequently, a vast literature in financial economics studies conglomerates, focusing on conglomerate value and internal capital allocation.¹ An important empirical challenge faced by this literature is that direct estimates of divisional valuations and investment opportunities are not available. In this

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Correspondence: Mikhail Simutin, Rotman School of Management, University of Toronto, 105 St. George Street, Toronto ON, Canada, M5S 3E6; e-mail: Mikhail.Simutin@Rotman.Utoronto.Ca.

¹ See, for example, Lang and Stulz (1994), Berger and Ofek (1995), Servaes (1996), Lins and Servaes (1999), Rajan, Servaes, and Zingales (2000), Campa and Kedia (2002), Denis, Denis, and Yost (2002), Graham, Lemmon, and Wolf (2002), Billett and Mauer (2003), Villalonga (2003, 2004a, 2004b), Xuan (2009), Ozbas and Scharfstein (2010), Matvos and Seru (2014), and Matvos, Seru, and Silva (2018).

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paper, we address this challenge by developing a new method to estimate divisional valuation multiples, or Tobin's q s. We use our estimates to investigate differences between conglomerates and standalone firms and their implications for asset reallocation inside conglomerates and across firms.

To understand our method, consider the traditional approach to studying conglomerates, which uses industry q s derived from standalone firms to approximate the q s of conglomerate divisions. This approach assumes that q s of standalone firms accurately proxy for those of conglomerate divisions, and thus, does not account for systematic differences between q s of conglomerates and standalone firms that may arise due to endogeneity of firms' organizational structure and links between the conglomerate units. In this spirit, Maksimovic and Phillips (2002) and Gomes and Livdan (2004) propose models with different q s for conglomerates and standalone firms.²

To overcome this limitation, we estimate conglomerate-implied industry q s by mimicking standalone firms with portfolios consisting only of conglomerates. As a simple example, consider two conglomerates that operate in the same two industries. Assume that the first conglomerate has unit exposure of fundamentals to each industry, while the second conglomerate has exposures of two and one to these industries. It is immediately clear that a portfolio long the second conglomerate and short the first conglomerate has industry exposures of one and zero. The value of this portfolio therefore equals the conglomerate-implied valuation of divisions operating in the first industry and provides a proxy for the Tobin's q of these divisions.

In this example, the solution to the problem is unique and can be obtained by inverting the matrix containing conglomerate exposure weights to each industry. In practical applications, the number of conglomerates exceeds the number of industries and the matrix of conglomerate exposure weights is not invertible. We show that median regressions of conglomerate q s on the matrix of exposure weights provide reliable estimates of the divisional q s.

Our estimates of divisional q s vary across industries and over time and, unlike traditional measures, are obtained without relying on information from standalone firms. These features allow us to investigate the systematic differences between the q s of conglomerate divisions and those of standalone firms that operate in the same industries, and to shed new light on the theory and operation of corporate diversification.

In our first set of analyses, we investigate the properties of divisional q s and compare them to traditional measures of q derived from standalone firms. We find large differences between the two. Across industries, the average ratio of divisional and standalone q s varies considerably, ranging from -56% to 16% . Relative to standalone firms, divisional q s are lowest in the energy, high-tech, and healthcare industries, whereas divisions in the consumer nondurables and

² Furthermore, Whited (2001) shows that accounting for measurement error in estimates of Tobin's q derived from standalone firms is important to study the investment of conglomerate divisions.

telecommunication industries have higher qs than their standalone counterparts. Over time, divisional qs are 17% less volatile compared to standalone qs .

Together, these findings have broad implications for prior research on conglomerates, as they reveal systematic differences between the qs of conglomerate divisions and those of standalone firms. Consequently, research that relies on qs of standalone firms, such as studies of internal capital allocation, may mischaracterize intraconglomerate policies. Moreover, conclusions from industry-specific studies may not generalize to other industries, where the relation between divisional and standalone qs can differ considerably.³ Finally, these findings provide direct evidence that standalone qs and divisional qs are materially different, consistent with the models of Maksimovic and Phillips (2002) and Gomes and Livdan (2004).

In our second set of analyses, we investigate the economic mechanisms that explain the differences between divisional and standalone qs across industries and over time. We find that divisional qs are considerably less sensitive to market-wide and industry-specific economic fluctuations than standalone qs . These findings hold across a wide range of economic shocks, measured, for example, by stock market returns, dividend yields, and changes in industrial production or aggregate output.

To understand these differences, we build on a recent literature that shows that internal capital reallocation mitigates adverse shocks from financial markets (e.g., Matvos and Seru (2014), Almeida, Kim, and Kim (2015), Kuppuswamy and Villalonga (2016), Matvos, Seru, and Silva (2018)). We construct several measures of credit shocks, such as changes in the TED spread, changes in average industry credit ratings, or credit default swap (CDS) spreads, and find that the sensitivity of divisional qs to credit shocks is, on average, 54% lower compared to standalone firms. This evidence suggests that conglomerate divisions are more insulated from external financing shocks than standalone firms operating in the same industry.

Moreover, in multivariate regressions, we find that credit shocks absorb the effects of other economic shocks, suggesting that credit conditions are a key factor behind the wedge between standalone and divisional qs . This evidence supports the important role of capital reallocation inside conglomerates in the presence of external financial frictions (e.g., Alchian (1969), Weston (1970), Stein (1997), Hadlock, Ryngaert, and Thomas (2001), Matsusaka and Nanda (2002), Matvos and Seru (2014)).

To further explore the role of internal capital markets and coinsurance in mitigating the effects of credit shocks, we exploit a feature of our method that allows us to estimate divisional qs not only at the industry level but also at the

³ Studies that focus on a single industry include Lamont (1997), who analyzes investment decisions of diversified oil companies following the oil price shock of 1986; Khanna and Tice (2001), who investigate the responses of diversified firms to Walmart's entry into their market; Campello (2002), who studies the reactions of financial conglomerates to monetary policy; and Guedj and Scharfstein (2004), who study biopharmaceutical firms. Additionally, a long line of research including Villalonga (2004a), Maksimovic and Phillips (2002), and Schoar (2002) uses information from establishment-level data available only for the manufacturing sector.

level of any observable division characteristic. We estimate qs across divisions that have low versus high covariances with the rest of the conglomerate and find that the sensitivity of qs to credit shocks in low-covariance divisions is insignificant. This evidence suggests that coinsurance plays an important role in insulating divisions from external financial frictions. Moreover, credit shocks absorb the effects of other economic shocks in explaining the differences between qs of high- and low-covariance divisions, indicating that the effects of covariance structures operate through a financing channel.

Collectively, these findings support theories of coinsurance and internal capital reallocation, which suggest that covariance structures play an important role in conglomerate financing and investment (e.g., Lewellen (1971), Matvos, Seru, and Silva (2018)). Broadly interpreted, the results show that differences between qs of standalone firms and conglomerate divisions reflect differences in their sensitivities to credit conditions and the importance of resource reallocation through internal capital markets inside conglomerates. Hence, our divisional qs should be important to understanding asset (re)allocation both inside conglomerates and across focused and diversified firms.

To investigate this hypothesis in our third set of analyses, we start by examining conglomerates' internal capital allocation following the empirical approach of Ozbas and Scharfstein (2010), whose study of the neoclassical relation between investment and Tobin's q builds on the early work of Shin and Stulz (1998). Using qs derived from standalone firms, they find that investment- q sensitivity is lower for conglomerate divisions than standalone firms.⁴

We confirm the findings of Ozbas and Scharfstein (2010): conglomerate divisions' elasticity of investment to standalone qs is lower than that of standalone firms. However, conglomerate investment is much more sensitive to divisional qs . While measuring investment efficiency is notoriously difficult, our results provide strikingly different estimates of investment- q sensitivities of conglomerate divisions and highlight the importance of divisional qs for internal capital allocation.

A possible concern with the analysis above is that the sensitivity of investment to divisional qs is correlated with potentially unobservable firm-level characteristics. While the analysis includes firm-division fixed effects that absorb time-invariant differences between standalone firms and divisions, we further address this concern by estimating the elasticity of investment to standalone and divisional qs around the spinoff of conglomerate divisions into standalone firms. This analysis holds constant the firm and traces its invest-

⁴ Existing theory offers diverging views on the efficacy of conglomerates' internal capital allocation. On the one hand, internal capital markets in conglomerates may allow firms to raise more external capital (Lewellen (1971), Hadlock, Ryngaert, and Thomas (2001)) and to invest more efficiently (Alchian (1969), Weston (1970), Stein (1997), Matsusaka and Nanda (2002)). On the other hand, conglomerates may suffer from agency problems such as the rent-seeking behavior of divisional managers. Consequently, the CEO may distort the conglomerate's internal capital allocation toward weaker divisions to retain divisional managers (Scharfstein and Stein (2000)) and to control their rent-seeking behavior (Rajan, Servaes, and Zingales (2000)).

ment policy over a short window surrounding the organizational changes that it undergoes, thereby minimizing confounding effects of time-varying unobservable differences across firms. The results suggest that prior to the spinoff, divisional investment is highly sensitive to divisional q and less sensitive to standalone q , whereas following the spinoff, investment becomes insensitive to divisional q and more sensitive to standalone q . These results suggest that the change in organizational structure following spinoffs and its implications for investment opportunities are reflected in divisional and standalone qs .

We next investigate resource allocation across focused and diversified firms by analyzing diversifying mergers and acquisitions. We find that the differences between divisional and standalone qs predict the number of diversifying acquisitions in industries as well as their announcement returns.

Taken together, our findings show that the differences between divisional and standalone qs explain asset allocation decisions both inside conglomerates and across conglomerates and focused firms. This result is consistent with our earlier evidence that frictions in external capital markets and access to internal capital markets give rise to a wedge between divisional and standalone qs .

Our study highlights that conglomerate divisions benefit from better access to capital markets, which may seem hard to reconcile with the “diversification discount” documented by Lang and Stulz (1994), Berger and Ofek (1995), and related literature. However, studies such as Campa and Kedia (2002), Graham, Lemmon, and Wolf (2002), and Villalonga (2004b) show that after accounting for firms’ endogenous decision to diversify, conglomerates no longer appear to be discounted. Our findings support this view by showing that shifts in divisional qs relative to standalone qs , which capture variation in the valuation and investment opportunities of conglomerate divisions versus standalone firms, explain asset allocations within and across firms.

Overall, our paper contributes to the literature on corporate diversification and internal capital markets by improving our understanding of the economic mechanisms that underlie the differences between standalone firms and conglomerates. However, our method is not without limitations. First, our approach estimates valuations from a group of potentially heterogeneous divisions from different conglomerates. Nonetheless, it represents a substantial improvement relative to traditional methods (e.g., Berger and Ofek (1995)) that additionally assume away differences between divisions and standalone firms. Second, our approach groups divisions from conglomerates with different, nonrandom compositions and synergy structures. While estimates based on standalone qs exclude synergies and cross-dependencies by design, our method does account for them by allocating them proportionally to the divisions’ industry sales weights. Finally, we caution the reader that our method and the resulting divisional qs do not overcome the standard identification challenge that arises when trying to study the causal effect of corporate diversification on firm value. Accordingly, our analyses are not intended to provide causal evidence on efficiency and value.

While we focus on external financing frictions, our method can be extended to study other economic mechanisms that might help explain the differences

between standalone firms and conglomerates, including cost structures, supplier networks, bargaining, competitiveness, innovation, and synergistic spillovers across product markets and supply chains.

The paper is organized as follows. In Section I, we illustrate our new method of estimating divisional qs without relying on standalone firms. Section II describes the data and the empirical approach. Section III investigates the characteristics and determinants of divisional qs . We study the real effects of divisional qs for internal and external capital allocation in Section IV. Finally, in Section V, we conclude.

I. The New Method

Our method inverts the traditional approach of Lang and Stulz (1994) and Berger and Ofek (1995). Rather than building up synthetic conglomerates from individual pieces (standalone firms), we break down actual conglomerates into their components. We group these components across conglomerates into classes that share observable characteristics, notably industry affiliation. We then use quantile regressions to obtain median conglomerate-implied estimates for the valuation multiples, or qs , of each class. Comparing these qs with those of standalone firms in each class allows us to analyze the systematic differences between conglomerates and standalone firms at the granular level of a single class.

Let W denote the matrix of I conglomerates by K classes that contains the fundamentals (e.g., sales) of a cross-section of conglomerates. For example, in the analysis of Berger and Ofek (1995), K represents the number of four-digit SIC industries in which the I conglomerates operate. If divisions in all classes function independently, the value of the conglomerate, v , equals the weighted sum of the valuation ratios of the classes represented by the $K \times 1$ vector q^C :

$$v = W \cdot q^C. \quad (1)$$

In the special case in which W contains the replacement costs of capital, q^C corresponds to the Tobin's q associated with each class. Because replacement costs are not observable, the corporate diversification literature relies on asset or sales multiples to proxy for Tobin's qs (Lang and Stulz (1994)).

The traditional approach of Lang and Stulz (1994) and Berger and Ofek (1995) imputes the value of conglomerates using industry-level qs estimated from standalone firms. In particular, they proxy for the q of each division by the median q of standalone firms operating in the same industry, $\hat{q}^C = q^{SA}$. A standard result in this literature is that the imputed values of conglomerates, $\hat{v} = W \cdot q^{SA}$, on average, exceed their market values, suggesting that diversified firms trade at a discount.

Our method also builds on equation (1), but we estimate q^C using only conglomerate-level information. Specifically, we scale equation (1) by the total fundamentals of each conglomerate to obtain

$$\tilde{v} = \tilde{W} \cdot q^C, \quad (2)$$

where $\tilde{v}(i) = v(i) / \sum_k W(i, k)$ and $\tilde{W}(i, k) = W(i, k) / \sum_k W(i, k)$ are valuation multiples and class weights of the conglomerates. We note that our method is not designed to value conglomerates relative to standalone firms—using imputed values based on q^{SA} is appropriate for that purpose. Instead, we aim to estimate qs of divisions operated by conglomerates to investigate the economic differences between those divisions and standalone firms.

Estimating $\hat{q}^c$ from equation (2) could in theory be achieved via an ordinary least squares (OLS) regression of conglomerate multiples on class weights. However, the OLS approach is problematic since valuation ratios are positively skewed. Addressing this skewness by taking natural logarithms is not suitable for our purposes since logs are not additive. To address the problem of skewed valuation ratios, we base our analysis on medians rather than means and use quantile regressions. Specifically, we run median regressions of conglomerate multiples $\tilde{v}$ on class weights $\tilde{W}$ cross-sectionally to back out the median class valuation ratios $\hat{q}^c$. The cross-sectional nature of the regression allows all variables to change each year. Therefore, $\hat{q}^c$ s vary in the time series in response to changes in both conglomerate values and fundamentals. In the [Internet Appendix](#), we provide more details on quantile regressions and we show in Monte Carlo simulations that divisional qs can be robustly estimated using median regressions rather than OLS if conglomerate valuation ratios are positively skewed.⁵

II. Data and Empirical Approach

A. Data

We obtain firm-level accounting variables and SIC industry classifications from Compustat, and division-level variables from the Compustat Segment files.⁶ Our sample period starts in 1978, when Compustat segment data become available, and ends in 2018. Following the literature, we exclude firms with at least one division in the financial sector (SIC codes 6000-6999), in agriculture (SIC codes below 1000), and in government, other noneconomic activities, or unclassified services (SIC codes 8600, 8800, 8900, and 9000).

Since we are interested in studying divisional investment opportunities, we exclude firms whose sales at the business segment level are unavailable on Compustat. We also exclude divisions with zero sales such as corporate accounts. Following the literature, we further require total sales from the

⁵ The [Internet Appendix](#) is available in the online version of this article on *The Journal of Finance* website.

⁶ By using Compustat data, our method calculates divisional qs based on conglomerate segments whose reporting can be discretionary. We note, however, that our divisional qs are estimated from median regressions that mitigate the effects of conglomerate-specific, idiosyncratic misreporting and outliers. Moreover, our method can be applied to other data sets that include establishment- or plant-level data that are less prone to these data limitations (e.g., (Maksimovic and Phillips (1998, 2001, 2002, 2007)), Schoar (2002), Villalonga (2004b)). We leave for future work investigation of whether our method generates different estimates of qs using these alternative data sources.

Table I
Summary Statistics

This table reports summary statistics for the 1978 to 2018 sample of conglomerates defined on the basis of the Fama-French 10-industry classification. Sales and assets are in 2018 CPI-adjusted dollars. The second-to-last row shows the number of unique conglomerates and divisions. The last row shows the number of conglomerate-year and division-year observations.

	Conglomerates		Divisions	
	Mean	Median	Mean	Median
Sales, billion	5.42	0.99	2.43	0.32
Assets, billion	5.78	1.02	2.60	0.35
CAPEX-to-sales ratio, percent	9.07	4.23	12.21	3.78
Value-to-sales ratio (q)	1.43	0.99		
Number of segments	2.29	2.00		
Number of unique observations		2,575		6,427
Pooled number of observations		13,033		29,809

Compustat annual files to be greater than \$20 million and within 1% of the sum of divisional sales.

While earlier literature considers both sales- and assets-based multiples, Custodio (2014) offers a caveat by noting that the accumulation of goodwill in merger and acquisition accounting biases the book value of assets of conglomerates upwards, and that conglomerates have more flexibility in allocating assets across divisions. We therefore base our analysis on sales multiples.

We define a conglomerate as a firm that operates in at least two distinct industries. The literature frequently classifies industries by four-digit SIC codes. A shortcoming of this definition is that the number of conglomerate divisions that operate in each industry in a given year is small, with a median number of divisions per industry-year of just two. Our subsequent empirical analyses therefore rely on the 10 Fama-French industries, which are sufficiently broad to ensure that a meaningful number of divisions operate in each industry in a given year and thus that the matrix of industry weights for conglomerates, $\bar{W}$, is well populated.⁷ A benefit of this broader classification is that our analysis is less exposed to the criticism of Villalonga (2004a), who highlights problems in the four-digit SIC assignment by Compustat. While a substantial number of divisions may be misclassified into the wrong four-digit SIC codes, they likely remain in the same Fama-French 10 industry.

Table I reports sample summary statistics, which includes 2,575 conglomerates (13,033 firm-year observations) and 6,427 divisions (29,809 division-year observations). In this sample, an average (median) conglomerate has annual sales of \$5.42 (\$0.99) billion Consumer Price Index (CPI)-adjusted 2018 dollars, owns book assets valued at \$5.78 (\$1.02) billion, has a Tobin's q of

⁷ The industry definitions are provided in Appendix A. The Internet Appendix shows that the main results are robust to using the Fama-French 17-, 30-, and 49-industry classifications.

1.43 (0.99) based on sales multiples, operates in 2.29 (2) business segments, and has annual capital expenditures of 9.07% (4.23%) of sales. An average (median) division has annual sales of \$2.43 (\$0.32) billion and owns book assets valued at \$2.60 (\$0.35) billion.

Requiring that conglomerates operate in two distinct industries based on the 10 Fama-French industry classification reduces the number of conglomerates by about one-third relative to the four-digit SIC -based definition. However, the characteristics of this smaller sample are consistent with those reported in the literature (e.g., Custodio (2014)).

Finally, we note that we continue to use the standard four-digit SIC industry classification in the construction of standalone qs when feasible, for example, in the investment- q sensitivity analysis. We do so to maintain the natural advantage of the traditional measures of industry qs . These measures can be readily estimated from a sufficient number of standalone firms in each four-digit SIC industry cross-section.

B. Empirical Approach

In this section, we describe the empirical approach to estimating divisional qs . Each year, we estimate cross-sectional quantile regressions, without intercept, of conglomerate qs on their sales exposure to the 10 Fama-French industries. The resulting coefficients are estimates of the median q of conglomerate divisions in each industry-year. The annual number of divisions in each industry ranges from 26 (Telecommunications) to 169 (Manufacturing) on average, which ensures that each regression coefficient is well identified.

This approach generates direct estimates of conglomerate divisions' qs . Since it does not rely on the qs of standalone firms, our method eliminates measurement errors that may arise due to systematic differences between conglomerates and standalone firms. This allows us to characterize the differences between the qs of conglomerate divisions and those of standalone firms. An identifying assumption, shared with the traditional approach (e.g., Lang and Stulz (1994), Berger and Ofek (1995)), is that divisions of different conglomerates in each class have similar valuation ratios.

Since there are more standalone firms than conglomerates, the traditional approach is typically applied at more granular industry levels. To assess whether our approach more accurately approximates the overall value of the conglomerate despite the coarser industry classification, we follow Berger and Ofek (1995) and define excess value as the log-difference between conglomerates' true and imputed values for each firm-year. Imputed values are calculated as the sales-weighted sum of either the median qs of standalone firms in the respective four-digit SIC industries or the new divisional qs obtained from quantile regressions using the 10 Fama-French industry classification. We then compute the time-series standard deviations of both excess value measures for each firm with at least five consecutive observations.

Figure 1 plots the kernel density of these standard deviations. The figure clearly shows that our method results in excess value estimates with a

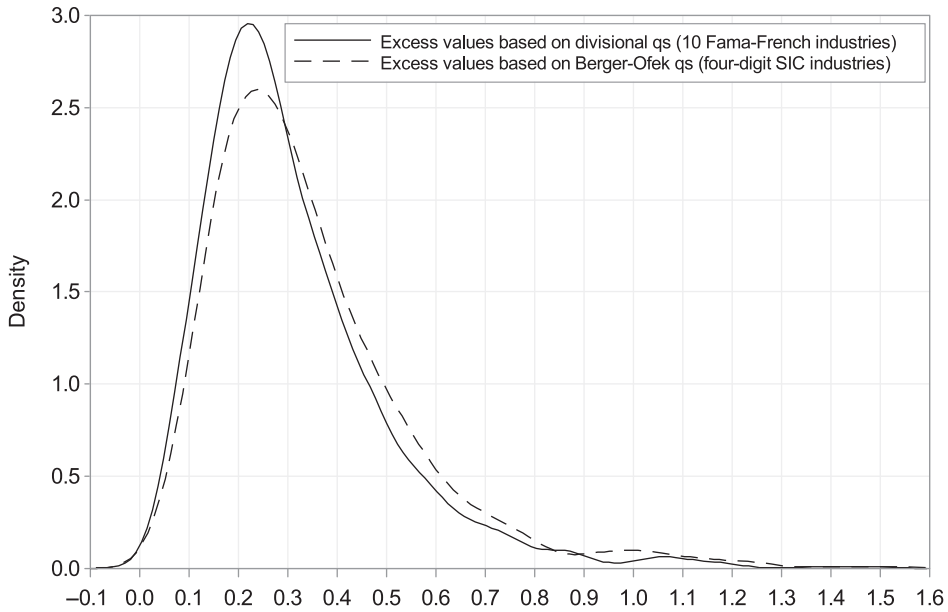


Figure 1. Kernel density of standard deviations of conglomerate excess values. The figure plots the kernel density of the time-series standard deviation of two measures of conglomerates' excess value, where excess value is the log-difference between the conglomerates' true and imputed values. Imputed values are calculated as the sales-weighted sum of either the new divisional qs obtained from quantile regressions using the 10 Fama-French industries (solid line) or the median qs of standalone firms in the respective four-digit SIC industries as in Berger and Ofek (1995) (dashed line). For each firm, the standard deviation of the timeseries of its excess values is computed, requiring at least five consecutive observations.

lower standard deviation than the estimates produced using the traditional approach of Berger and Ofek (1995). This evidence suggests that, despite being defined at a coarser industry level, the new estimates of intra-conglomerate qs more closely approximate the true conglomerate value than the standard industry-median qs of standalone firms.

In the remainder of the paper, we characterize the new measures of divisional qs and investigate the economic differences between conglomerates and standalone firms. The analyses focus on external financing frictions and the allocation of capital and assets both within firms, through internal capital markets, and across firms, through diversifying acquisitions and spinoffs.

III. The Characteristics and Determinants of Divisional qs

A. Cross-Industry Analysis

We begin our analysis by presenting evidence on the average valuation multiples, or qs , of conglomerate divisions and standalone firms across industries. Our focus on industry-level valuation estimates is motivated by the standard

definition of a diversified firm in corporate finance—a firm that operates in more than one industry. As suggested by Coase (1937) and Maksimovic and Phillips (2007), corporate diversification matters only if the conglomerate's industry composition affects its transaction costs and hence its optimal boundaries. Thus, for corporate diversification to be of interest, it must be the case that the industry composition of conglomerates is related to firm value.

In Table II, for each of the 10 Fama-French industries, we estimate average valuation multiples of conglomerate divisions and standalone firms, as well as their log difference (relative divisional qs , RDQs). The results offer three main takeaways. First, the valuation multiples of conglomerate divisions are significantly different from those of standalone firms. The time-series correlation between the qs of divisions and standalone firms ranges from 0.40 in the health sector to 0.96 in manufacturing. Moreover, the RDQs are statistically significantly different from zero in all sectors except manufacturing.⁸ To our knowledge, this is the first direct evidence that standalone qs and divisional qs are materially different, corroborating the assumptions and implications of existing models of corporate diversification such as Maksimovic and Phillips (2002) and Gomes and Livdan (2004). These estimates also suggest that industry multiples of standalone firms are noisy proxies for divisional multiples and can introduce measurement errors. This evidence is consistent with Whited (2001), who shows that accounting for measurement error in estimates of Tobin's qs derived from standalone firms is important for the study of investment by conglomerate divisions.

Second, Table II shows that the valuation multiples of divisions relative to standalone firms vary considerably across industries. Seven industries have significantly negative RDQs, with the largest discounts in the energy (−56%), high-tech (−33%), and healthcare (−31%) industries. In contrast, conglomerate divisions in two industries show substantial premia: divisions in the nondurable goods sector are valued at a 12% premium and divisions in the telecommunications sector are valued at a 16% premium. Divisions in the manufacturing sector exhibit neither a discount nor a premium, with a point estimate of 0.00 for its RDQ and a t -statistic of 0.04.

Third, Table II reveals considerable within-industry variation in RDQs over time. The standard deviation in RDQs within industries ranges from 11% and 12% in the energy and manufacturing sectors to 38% in the healthcare sector. This variation is illustrated in Figure 2, which shows the time series of RDQs for each industry. These estimates suggest that the relative values of diversified versus focused organizational structures within each industry are not constant over time.

Overall, our estimates indicate that qs of conglomerates relative to standalone firms vary systematically across industries and over time. These findings have broad implications for research on corporate diversification. First,

⁸ To establish the statistical properties of the time-series averages of RDQs, each year we re-sample residuals of the quantile regression 1,000 times and reestimate the regression. We then compute average RDQs for each bootstrapped sample and calculate their standard deviation.

Table II
Industry Analysis of Standalone and Divisional *qs*

This table reports *qs* of conglomerate divisions and standalone firms, as well as their log difference (relative divisional *qs*, RDQs) for each of the 10 Fama-French industries. The table also reports bootstrapped *t*-statistics and time-series moments.

	NDur	Dur	Manu	Enrg	HiTec	Tlcm	Shop	Hlth	Util	Oth
Divisional <i>q</i>	0.92	0.64	0.90	1.37	1.19	3.21	0.43	1.45	1.97	1.11
Std. deviation	0.34	0.21	0.33	0.44	0.68	0.85	0.16	0.60	0.79	0.42
Standalone <i>q</i>	0.87	0.74	0.88	3.15	1.72	2.88	0.51	2.49	2.24	1.30
Std. deviation	0.39	0.18	0.24	0.85	0.64	0.95	0.13	1.30	0.64	0.49
Correlation	0.56	0.67	0.96	0.63	0.83	0.68	0.57	0.40	0.91	0.58
RDQ	0.12	-0.13	0.00	-0.56	-0.33	0.16	-0.16	-0.31	-0.14	-0.10
<i>t</i> -statistic	[3.07]	[-2.65]	[0.04]	[-37.6]	[-17.7]	[5.68]	[-3.47]	[-14.0]	[-6.02]	[-4.66]
Std. deviation	0.30	0.19	0.12	0.11	0.17	0.28	0.23	0.38	0.17	0.30
RDQ, percentile 25	-0.03	-0.26	-0.09	-0.64	-0.46	-0.06	-0.35	-0.59	-0.26	-0.31
RDQ, median	0.14	-0.16	-0.03	-0.56	-0.36	0.11	-0.18	-0.36	-0.13	-0.17
RDQ, percentile 75	0.28	-0.08	0.11	-0.50	-0.23	0.31	0.00	-0.15	-0.01	0.09
No. conglomerates	64	53	169	51	84	26	98	29	35	117

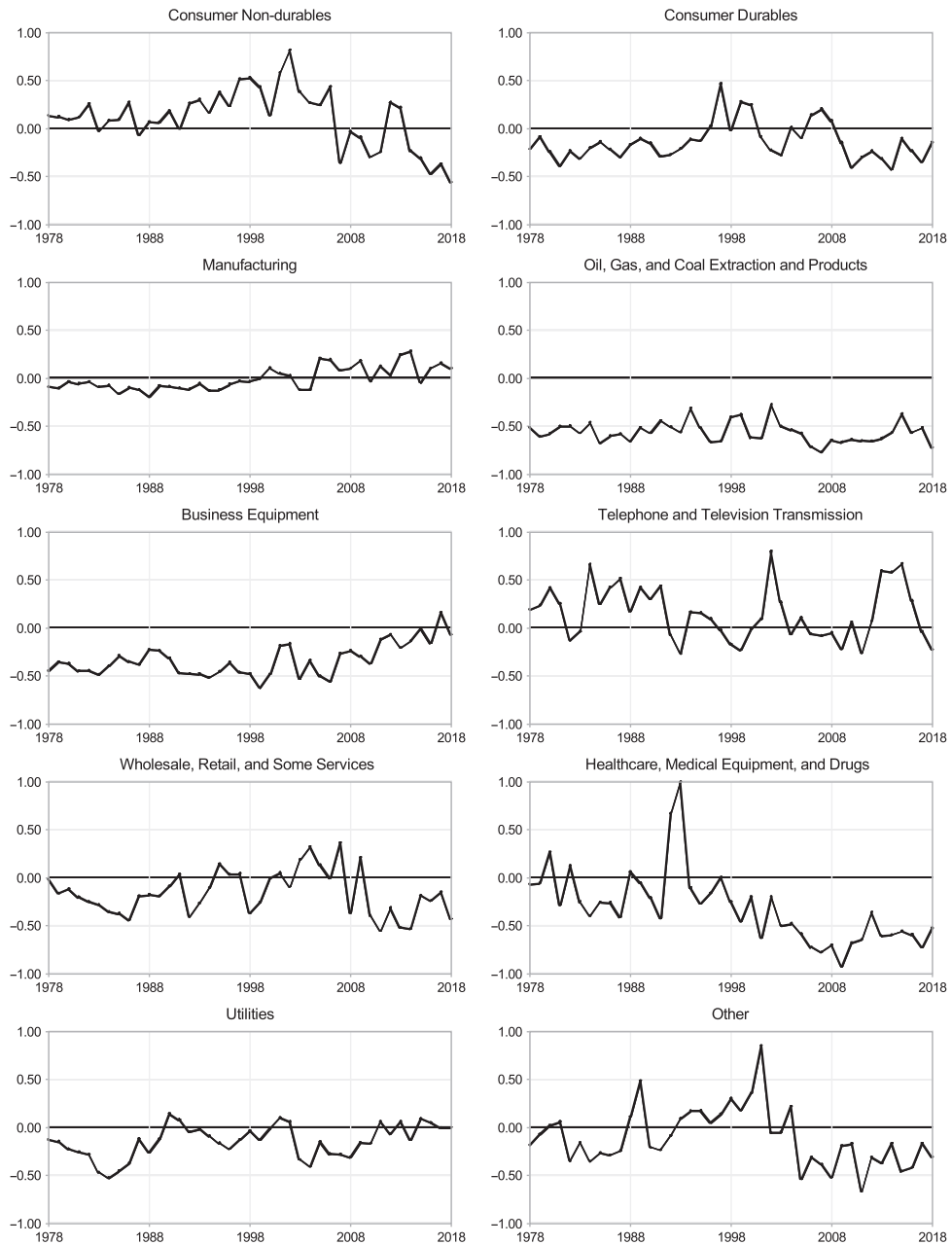


Figure 2. The timeseries of relative divisional q_s . This figure plots relative divisional q_s for divisions in the 10 Fama-French industries.

our study suggests that focusing on the systematic heterogeneity in conglomerate valuations across industries is important in studying corporate diversification. Most prior research does not directly account for this dimension (e.g., Rajan, Servaes, and Zingales (2000), Scharfstein and Stein (2000), Graham, Lemmon, and Wolf (2002), Campa and Kedia (2002), Villalonga (2004a, 2004b)). The variation in RDQs across industries also suggests that one needs to exercise caution in extrapolating results based on industry-specific analyses. For example, several studies use the Longitudinal Research Database to investigate the value of internal capital markets (e.g., Maksimovic and Phillips (2002)) and the productivity of conglomerate divisions (Schoar (2002)). However, this database tracks only manufacturing plants, a limitation that the authors of these studies acknowledge. Our estimates confirm that there is no significant discount in the manufacturing sector, but also uncover deep discounts or premia in other sectors.

Second, the variation in industry RDQs over time suggests that the trade-offs between diversified and focused organizational structures also vary with changing industry conditions. Prior studies of corporate diversification mostly focus on time-invariant, average differences in the value of diversified and focused firms. Notable exceptions include Matsusaka (1993) and Matvos, Seru, and Silva (2018), who study aggregate trends in corporate diversification. Our findings highlight the importance of studying disaggregated, industry-level trends in corporate diversification.

One caveat with the analysis thus far is that a conglomerate's choice of industries is nonrandom. For example, Hoberg and Phillips (2018) show that conglomerates tend to operate in economically related industries. We consider conglomerates' endogenous industry matching by studying within-conglomerate cross-industry pairs. Each column in Panel A of Table III corresponds to all conglomerates operating a division in the given industry, and each row reports the proportion of conglomerates that also have divisions in the industry indicated by the row. The sum of each column exceeds one because conglomerates can operate in more than two industries. Consistent with nonrandom industry matching, we find that industry pairs are not equally distributed inside conglomerates. This can be seen from the nonuniform distribution within each row. For example, 51% of conglomerates with a division in utilities also operate a division in energy, whereas only 4% of conglomerates with a division in nondurables do so.

To assess the effect of endogenous industry matching on our q estimates, Panel B of Table III presents average RDQs obtained by omitting from the sample all conglomerates that operate in the industry indicated in the row label. For example, the first row indicates that when the sample is limited to conglomerates that do not operate in the nondurables sector, divisions in the durables industry have estimated q s that are 11% smaller than those of standalone firms. Across all industries, the exclusion of industry pairs does not have a sizeable effect on the average RDQ. These exclusions generate small variation in RDQs within each industry, with standard deviations ranging from 2% to 7%. Furthermore, the RDQs estimated after excluding industry

Table III
Industry Analysis of Divisional *qs*: The Role of Industry Pairings

This table shows the conditional frequency distribution of industry pairings in conglomerates (Panel A) and relative divisional *qs* (RDQs) estimated when industry pairings are omitted (B). For conglomerates operating in the industry indicated by the column, Panel A reports the proportion that also have divisions in the industry indicated by the row. Panel B shows average RDQs obtained by omitting from the sample all conglomerates that operate in the industry indicated by the row. *, **, and *** indicate RDQs that are significantly different from those estimated in the full sample (Table II) at the 10%, 5%, and 1% level, respectively. The bottom four rows report the average, standard deviation, minimum, and maximum of these RDQs.

	NDur	Dur	Manu	Enrg	HiTec	Tlcm	Shop	Hlth	Util	Oth
Panel A. Conditional Distribution of Industry Pairings										
NDur	1.00	0.13	0.16	0.05	0.10	0.35	0.23	0.13	0.03	0.12
Dur	0.11	1.00	0.22	0.06	0.17	0.06	0.06	0.10	0.05	0.08
Manu	0.42	0.69	1.00	0.39	0.55	0.12	0.35	0.48	0.16	0.38
Enrg	0.04	0.06	0.12	1.00	0.04	0.07	0.13	0.06	0.51	0.17
HiTec	0.13	0.27	0.27	0.07	1.00	0.26	0.13	0.28	0.05	0.15
Tlcm	0.14	0.03	0.02	0.03	0.08	1.00	0.05	0.03	0.08	0.10
Shop	0.35	0.11	0.20	0.25	0.15	0.18	1.00	0.20	0.25	0.26
Hlth	0.06	0.05	0.08	0.04	0.10	0.04	0.06	1.00	0.03	0.04
Util	0.02	0.04	0.03	0.35	0.02	0.11	0.09	0.03	1.00	0.13
Oth	0.22	0.17	0.27	0.39	0.21	0.44	0.31	0.17	0.44	1.00
Panel B. Relative Divisional <i>qs</i> If Industry Pairings Are Omitted										
NDur		-0.11	0.00	-0.55	-0.33	0.09	-0.10	-0.31	-0.13	-0.10
Dur	0.12		-0.04	-0.55	-0.33	0.14	-0.16	-0.30	-0.14	-0.06
Manu	0.14	-0.24		-0.50**	-0.30	0.16	-0.15	-0.40*	-0.15	-0.01**
Enrg	0.13	-0.12	-0.02		-0.32	0.15	-0.15	-0.33	-0.05***	-0.14
HiTec	0.08	-0.09	0.00	-0.57		0.26**	-0.15	-0.28	-0.13	-0.04*
Tlcm	0.02	-0.13	0.01	-0.56	-0.32		-0.13	-0.30	-0.16	-0.13
Shop	0.15	-0.16	0.06*	-0.55	-0.32	0.22		-0.26	-0.05***	-0.13
Hlth	0.09	-0.11	-0.01	-0.55	-0.36	0.17	-0.16		-0.15	-0.08
Util	0.12	-0.13	0.02	-0.57	-0.32	0.15	-0.19	-0.31		-0.09
Oth	0.11	-0.14	0.01	-0.56	-0.34	0.00***	-0.13	-0.28	-0.19	
Average	0.11	-0.14	0.00	-0.55	-0.33	0.15	-0.15	-0.31	-0.13	-0.09
St. Dev.	0.04	0.04	0.03	0.02	0.02	0.07	0.02	0.04	0.05	0.04
Min	0.02	-0.24	-0.04	-0.57	-0.36	0.00	-0.19	-0.40	-0.19	-0.14
Max	0.15	-0.09	0.06	-0.50	-0.30	0.26	-0.10	-0.26	-0.05	-0.01

pairs are statistically different from the full-sample RDQs at the 10% level in nine out of 90 cases, or exactly 10% of cases. Overall, this evidence suggests that the endogenous within-conglomerate industry matching does not have a material effect on our estimates.

Taken together, the results in this section suggest that conglomerate-implied qs vary systematically over time and across industries and are substantially different from the commonly used industry qs based on standalone firms. In the next subsection, we investigate the economic mechanisms that explain the differences between the qs of standalone firms and conglomerate divisions and their variation across industries and over time.

B. Economic Determinants

We study the economic mechanisms that explain the variation in the qs of conglomerate divisions and in those of standalone firms by analyzing their sensitivity to a wide array of economic shocks. To explain the variation across industries and over time, we consider both aggregate and industry-specific shocks. These shocks include stock returns at both the overall market and the industry levels, changes in dividend yields at the market and industry levels, changes in the VIX, changes in industrial production, shocks to total factor productivity (TFP) and aggregate output, and an indicator equal to one for economic expansions. Data sources and construction details are provided in Appendix B. All variables are signed to have higher values when economic conditions improve.

In Table IV, for each of the 10 Fama-French industries, we regress annual changes in qs on contemporaneous economic shocks. We estimate these analyses separately for the qs of standalone firms (columns (1) and (2)), the qs of conglomerate divisions (columns (3) and (4)), and the differences between the qs of conglomerate divisions and standalone firms (columns (5) and (6)). Table IV reports the average slope coefficients and R^2 s across industries.

Panel A shows that the sensitivity of standalone qs to economic shocks is considerably higher than that of divisional qs across all measures, with the difference statistically significant in all cases. For example, a positive shock to industrial production is associated with increases in both standalone and divisional qs . However, the increase in standalone qs is much higher (6.24) than that in divisional qs (3.23), with a difference of 3.01 ($t = 2.92$). On average, across all of the economic shock measures, the sensitivity of divisional qs is a staggering 50% lower.

To reduce dimensionality, we aggregate the economic shocks using the first principal component of the variables in Panel A. We construct two versions of this aggregate time series. The first version is based only on those variables available over the entire sample period. The second version is based on an average first principal component that uses all combinations of available variables to maximize the estimation period.

The results in Panel B show that divisional qs are less sensitive to aggregate economic conditions than standalone qs . In particular, the sensitivity of

Table IV
Economic Determinants of Standalone and Divisional qs

This table investigates the sensitivity of standalone and divisional qs to economic conditions. It reports average slope coefficients and R^2 s from industry-level univariate regressions. The dependent variable is the annual change in q , normalized to have unit standard deviation, and the independent variable is a contemporaneous economic shock. In Panel B, the independent variables are two versions of the first principal component of the variables in Panel A based on either the variables available for the entire sample or the average first principal component that uses all combinations of available variables to maximize the estimation period. t -Statistics are reported in brackets. Data sources and construction details are provided in Appendix B. The sample period is 1979 to 2018, except for ΔVIX (1987 to 2018).

	Standalone qs		Divisional qs		Div. – SA q	
	Slope	R^2	Slope	R^2	Slope	R^2
Panel A. Economic Variables						
Market return	2.98 [13.34]	0.27	1.59 [4.07]	0.11	–1.39 [–4.96]	0.06
Industry return	2.44 [12.14]	0.39	1.03 [2.62]	0.12	–1.41 [–6.61]	0.13
– Δ Dividend yield	0.66 [6.39]	0.08	0.28 [3.38]	0.02	–0.37 [–3.74]	0.03
– Δ Industry dividend yield	0.42 [3.04]	0.04	0.21 [2.44]	0.01	–0.21 [–2.55]	0.02
– ΔVIX	0.09 [10.78]	0.20	0.05 [4.10]	0.09	–0.04 [–5.16]	0.03
Δ Industrial production	6.24 [8.77]	0.07	3.23 [2.91]	0.04	–3.01 [–2.92]	0.03
ΔTFP	0.21 [7.27]	0.11	0.09 [2.92]	0.03	–0.12 [–7.30]	0.04
Δ Output	0.08 [6.54]	0.05	0.04 [2.68]	0.02	–0.04 [–2.68]	0.02
Δ Expansion indicator	1.10 [9.30]	0.14	0.61 [5.87]	0.05	–0.49 [–3.59]	0.04
Panel B. First Principal Component of Economic Variables						
Full-sample variables	0.21 [11.93]	0.19	0.11 [4.50]	0.06	–0.11 [–5.82]	0.05
All variables, PC average	0.22 [12.48]	0.22	0.11 [4.44]	0.07	–0.11 [–6.16]	0.06

divisional qs to both principal component-based measures of economic shocks is 0.11, compared to sensitivities of 0.21 and 0.22 for standalone qs . The differences between the sensitivities of divisional and standalone qs remain highly statistically significant.

Interpreted broadly, these findings indicate that a key economic distinction between conglomerate divisions and standalone firms is that divisions are more insulated from external economic fluctuations. Consistent with these findings, we find that the average time-series standard deviation of divisional

qs , calculated across all industries, is 17% lower than that of standalone qs . This difference is economically large and statistically significant in a joint test (untabulated, $p = 0.00$).⁹

C. Financing Conditions

To investigate the channels underlying the above differences between qs of conglomerate divisions and those of standalone firms, we build on prominent theories of corporate diversification highlighting the role of internal capital markets. In particular, these theories posit that conglomerates' internal capital markets allow them to raise more external financing than their standalone counterparts (Lewellen (1971), Hadlock, Ryngaert, and Thomas (2001)) and to reallocate capital internally across different lines of business more efficiently (Alchian (1969), Weston (1970), Stein (1997), Matsusaka and Nanda (2002)). These theories provide a financing-based foundation for differences between standalone firms and conglomerate divisions. More recent related work by Matvos and Seru (2014), Almeida, Kim, and Kim (2015), Kuppuswamy and Villalonga (2016), and Matvos, Seru, and Silva (2018) confirms that external financing frictions play an important role in conglomeration.

To assess the effect of external financing frictions on the differences between conglomerate and standalone qs , we construct measures of credit shocks based on changes in the following variables: the TED spread (Matvos, Seru, and Silva (2018)), the default spread, industry-average credit ratings, industry-level credit spreads, the Merton (1974) distance to default, and CDS spreads. As before, data sources and construction details are provided in Appendix B, and all variables are signed to have higher values when financing conditions improve.

Table V reports evidence on the sensitivity of qs to credit shocks, with its methods and layout mimicking those in Table IV. The estimates in Panel A suggest that qs of conglomerate divisions are significantly less sensitive to credit shocks than those of standalone firms. These results hold across all measures we consider, with divisions' sensitivities on average more than 50% lower. These findings suggest that shocks to credit markets are an important determinant of the differences between the qs of conglomerate divisions and standalone firms.

In Panel B of Table V, we construct two versions of the first principle components of the credit shock variables following the approach used in Table IV. As only default spread and distance to default are available for the full sample, the first version is based on these two variables. The results echo the findings in Panel A. The sensitivity of divisional qs to credit shocks is around

⁹ This estimate likely understates the difference in standard deviations between divisional and standalone qs since the volatility of an estimated series is inflated by the average volatility of the noise in the estimation. While this affects both divisional and standalone qs , medians are generally estimated with less noise than quantile regression coefficients. When we manually adjust the time-series volatility estimates by subtracting the average cross-sectional estimation noise from bootstrapped samples, the standard deviation of divisional qs is 20% lower than that of standalone qs .

Table V
Financing Conditions: Standalone and Divisional *qs*

This table investigates the sensitivity of standalone and divisional *qs* to financing conditions. It reports average slope coefficients and *R*²s from industry-level univariate regressions. The dependent variable is the annual change in *q*, normalized to have unit standard deviation, and the independent variable is a contemporaneous credit shock. In Panel B, the independent variables are two versions of the first principal component of the variables in Panel A based on either the variables available for the entire sample or the average first principal component that uses all combinations of available variables to maximize the estimation period. *t*-Statistics are reported in brackets. Data sources and construction details are provided in Appendix B. The sample period is 1979 to 2018, except for ΔTED (1987 to 2018), ΔCredit rating (1986 to 2017), ΔCredit spread (2005 to 2016), and ΔCDS (1991 to 2018).

	Standalone <i>qs</i>		Divisional <i>qs</i>		Div. – SA <i>q</i>	
	Slope	<i>R</i> ²	Slope	<i>R</i> ²	Slope	<i>R</i> ²
Panel A. Credit Shock Variables						
–ΔTED spread	1.02 [4.92]	0.11	0.67 [4.77]	0.04	–0.35 [–2.42]	0.02
–ΔDefault spread	0.53 [7.12]	0.10	0.23 [2.17]	0.05	–0.30 [–2.89]	0.05
–ΔCredit rating	0.51 [3.06]	0.04	0.23 [1.56]	0.03	–0.27 [–2.44]	0.01
–ΔCredit spread	0.43 [6.14]	0.44	0.14 [1.11]	0.19	–0.29 [–2.09]	0.17
ΔDistance to default	0.37 [9.47]	0.37	0.18 [3.04]	0.14	–0.19 [–6.98]	0.11
–ΔCDS	0.56 [3.71]	0.35	0.23 [2.00]	0.13	–0.33 [–2.62]	0.11
Panel B. First Principal Component of Credit Shock Variables						
Full-sample variables	0.46 [10.92]	0.32	0.21 [2.97]	0.12	–0.26 [–5.35]	0.10
All variables, PC average	0.44 [10.51]	0.33	0.20 [3.21]	0.12	–0.24 [–5.73]	0.09

0.20, whereas it is approximately 0.45 for standalone *qs* and the difference is highly statistically significant. Overall, the results in Table V indicate that conglomerate divisions are more insulated from external financing shocks than standalone firms.

In Table VI, we investigate the channel responsible for this insulation. Specifically, we estimate the sensitivity of *qs* to credit shocks for divisions that differ by their covariance with the rest of the conglomerate. Theory suggests that covariances play an important role, as lower values increase the scope for coinsurance and internal capital reallocation to mitigate the effects of external financing frictions and adverse shocks to credit supply (Lewellen (1971), Matsusaka and Nanda (2002), Matvos, Seru, and Silva (2018)).

To estimate this coinsurance potential, we first calculate the full-sample time-series covariance of annual sales growth rates of standalone firms for

Table VI
Financing Conditions: The Role of Coinsurance

This table investigates the sensitivity of qs of divisions with low and high coinsurance potential to financing conditions. It reports average slope coefficients and R^2 s from industry-level univariate regressions. To estimate qs , conglomerate divisions are grouped by both their Fama-French 10-industry affiliation and their coinsurance potential, which is inversely related to a division's relative contribution to the conglomerate variance. See Section III.C in the text for more details. The dependent variable is the annual change in q , normalized to have unit standard deviation, and the independent variable is a contemporaneous credit shock. In Panel B, the independent variables are two versions of the first principal component of the variables in Panel A based on either the variables available for the entire sample or the average first principal component that uses all combinations of available variables to maximize the estimation period. t -Statistics are reported in brackets. Data sources and construction details are provided in Appendix B. The sample period is 1979 to 2018, except for Δ TED (1987 to 2018), Δ Credit rating (1986 to 2017), Δ Credit spread (2005 to 2016), and Δ CDS (1991 to 2018).

	Divisional q When Coinsurance Potential Is					
	Low		High		High – Low	
	Slope	R^2	Slope	R^2	Slope	R^2
Panel A. Credit Shock Variables						
– Δ TED spread	0.79 [4.59]	0.06	0.15 [1.01]	0.02	–0.64 [–2.50]	0.03
– Δ Default spread	0.28 [2.39]	0.07	0.03 [0.25]	0.05	–0.25 [–1.76]	0.04
– Δ Credit rating	0.62 [3.82]	0.04	–0.33 [–1.90]	0.02	–0.95 [–3.64]	0.04
– Δ Credit spread	0.15 [1.70]	0.17	–0.09 [–0.72]	0.10	–0.24 [–2.68]	0.11
Δ Distance to default	0.21 [3.84]	0.17	0.03 [0.95]	0.03	–0.18 [–3.27]	0.07
– Δ CDS	0.33 [2.44]	0.18	–0.01 [–0.10]	0.06	–0.34 [–2.91]	0.09
Panel B. First Principal Component of Credit Shock Variables						
Full-sample variables	0.25 [3.50]	0.15	0.03 [0.55]	0.05	–0.22 [–2.56]	0.08
All variables, PC average	0.25 [3.73]	0.16	0.03 [0.53]	0.04	–0.22 [–2.63]	0.08

each three-digit SIC industry pair. Matching these estimates to divisions, we construct the sales-weighted covariance matrix for each conglomerate-year. We then calculate the relative contribution of a single division to the conglomerate's variance by dividing the sum of all matrix elements that are related to this division by the sum of all elements. High coinsurance divisions are defined as those with below-median relative contribution to conglomerate variance.

We estimate divisional qs as in equation (2) with $K = 20$ classes based on the bivariate coinsurance grouping within each 10 Fama-French industry. Note that the feasibility of such granular division-level analysis is a distinctive

Table VII
Sensitivity of qs to Financing and Other Economic Conditions

This table investigates the sensitivity of standalone and divisional qs (Panel A) as well as qs of divisions with low and high coinsurance potential (Panel B) to economic and financing conditions. It reports average slope coefficients and R^2 s from industry-level regressions of the annual change in q , normalized to have unit standard deviation, on contemporaneous economic and credit shocks, measured as in Panel B of Tables IV and V, respectively. In Panel A, qs are estimated for the 10 Fama-French industries. In Panel B, conglomerate divisions are grouped by both their Fama-French 10-industry affiliation and their coinsurance potential, as in Table VI. t -Statistics are reported in brackets. The sample period is 1979 to 2018.

Panel A. Standalone versus Divisional qs									
	Standalone qs			Divisional qs			Div. – SA q		
	Econ	Credit	R^2	Econ	Credit	R^2	Econ	Credit	R^2
Full-sample variables	0.06 [2.12]	0.40 [6.56]	0.34	0.04 [1.72]	0.16 [1.92]	0.13	−0.02 [−0.65]	−0.24 [−3.87]	0.11
All variables, PC average	0.06 [2.09]	0.38 [5.95]	0.36	0.03 [1.52]	0.17 [2.18]	0.14	−0.03 [−0.97]	−0.21 [−3.77]	0.11
Panel B. Divisional qs by Coinsurance Potential									
	Low			High			High – Low		
	Econ	Credit	R^2	Econ	Credit	R^2	Econ	Credit	R^2
Full-sample variables	0.03 [1.51]	0.21 [2.61]	0.17	0.01 [0.59]	0.02 [0.35]	0.06	−0.02 [−0.68]	−0.19 [−2.46]	0.09
All variables, PC average	0.03 [1.27]	0.21 [2.75]	0.18	0.01 [0.47]	0.02 [0.37]	0.06	−0.02 [−0.56]	−0.19 [−2.30]	0.10

feature of our method and would not be viable using traditional approaches based on the overall conglomerate.

Panel A of Table VI shows that, consistent with theory, coinsurance potential has a considerable impact on the sensitivity of divisional qs to credit shocks. While qs of low coinsurance divisions are sensitive to credit shocks, those of high coinsurance divisions are not. The results hold across the different credit shock measures, with the differences in sensitivities economically large and statistically significant. For the principal components in Panel B, we find a sensitivity of 0.25 ($t = 3.50$) for divisions with low coinsurance potential and an insignificant sensitivity of 0.03 ($t = 0.55$) for those with high coinsurance potential. These findings suggest that external financing frictions play an important role in divisional qs through internal capital reallocation across unrelated divisions.

In a final step, we evaluate the relative importance of financing and other economic shocks in explaining the differences in qs . In Table VII, we regress the annual changes in industry-level qs on the first principal component of

economic shocks and the first principal component of credit shocks. Panel A compares standalone firms and conglomerate divisions, while Panel B focuses on divisions with high versus low coinsurance potential.

The results in Table VII are striking. Panel A shows that after including credit shocks, other economic shocks do not help to explain the differences in the sensitivities of divisional and standalone q_s . In contrast, credit shocks remain significant. In particular, the coefficients on the credit shock variables, -0.24 and -0.21 , are little changed relative to the univariate specifications in Panel B of Table V (-0.26 and -0.24). In contrast, the coefficients on the other economic shocks (-0.02 and -0.03) are much smaller in magnitude than the corresponding values in Panel B of Table IV (-0.11 and -0.11). These results are all the more remarkable since the principal component for full-sample credit shocks is based on only two variables and thus is arguably noisier than that for economic shocks, which is based on eight variables.

Similar results obtain in Panel B, where credit shocks dominate the other measures of economic shocks in explaining the differences in the sensitivities of q_s of divisions with high versus low coinsurance potential. The coefficients on credit shocks are statistically significant at -0.19 , whereas the coefficients on the other economic shocks are statistically insignificant at -0.02 .

Taken together, the results in Tables IV through VII suggest that the q_s of standalone firms and conglomerate divisions differ considerably in their sensitivities to economic shocks. Conglomerate divisions are significantly more insulated from external economic shocks than standalone firms, and these differences operate through a credit channel whereby low-covariance structures across divisions mitigate their economic sensitivity through the reallocation of capital via internal capital markets.

Lastly, we note that while our analyses focus on credit shocks, our method of generating divisional q_s can be readily extended to study other, nonmutually exclusive economic mechanisms that might contribute to the differences between standalone firms and conglomerate divisions. Possible mechanisms include differences in investment cost structures, supplier networks, competitiveness, innovation, and synergistic spillovers across product markets and along the supply chain.

IV. Real Effects

The results in the previous section indicate that q_s of conglomerate divisions differ considerably from those of standalone firms, and that these differences originate from the ability of conglomerates to reallocate capital internally to mitigate the effects of external financing frictions. Consequently, the differences between divisional and standalone q_s should have real implications for capital reallocation both inside conglomerates through internal capital markets and across focused and diversified firms through mergers and acquisitions. Hence, in this section, we explore the real effects of divisional q_s for within-firm and across-firm investments, including analyses of internal capital allocation, spinoffs, and diversifying acquisitions.

A. Internal Capital Allocation

Neoclassical theory suggests that, absent financing frictions, investment should depend only on investment opportunities measured by marginal Tobin's qs . Estimation of marginal qs has been the focus of extensive research, which acknowledges that such estimation is particularly challenging for conglomerate divisions. We now use our estimates of divisional qs to revisit the evidence on the investment- q sensitivity of conglomerate divisions.

More concretely, Shin and Stulz (1998) and Ozbas and Scharfstein (2010) use the neoclassical relation between investment and q to study conglomerates' internal capital allocation. Both papers proxy for a division's q using the median q of standalone firms in its industry. Ozbas and Scharfstein (2010) compare the sensitivity of investment to Tobin's q in conglomerate divisions and standalone firms and find lower sensitivity for conglomerates. They interpret their results as evidence that conglomerates overinvest when opportunities are low and underinvest when they are high.

One concern with these studies is their use of standalone firms to proxy for the qs of conglomerate divisions. As argued by Whited (2001), Maksimovic and Phillips (2002), and Gomes and Livdan (2004), estimates of investment opportunities derived from the qs of standalone firms are inappropriate for the study of investment by conglomerate divisions. In contrast, our divisional qs do not rely on standalone firms and therefore are more suitable for examining the q -sensitivity of conglomerate divisions' investment.

Table VIII summarizes the results of our investment- q sensitivity analysis. As in Ozbas and Scharfstein (2010), the dependent variable is the annual ratio of capital expenditures to sales in each conglomerate division and standalone firm. Specifications (1) to (4) include year and industry fixed effects. Regressions (5) to (8) use year and firm-division fixed effects that absorb time-invariant unobservable differences across divisions.

The key explanatory variables are standalone and divisional qs and their interactions with C , an indicator variable equal to one for conglomerate divisions and zero for standalone firms. Standalone qs are estimated at the granular four-digit SIC level, whereas our divisional qs are available for the coarser Fama-French 10 industries.¹⁰ Whether our method's advantage of using estimates obtained without information from standalone firms outweighs the potential loss of information from the coarser industry classification is an empirical question.

Regressions (1) and (2) study the sensitivity of investment to each measure of q separately. The sensitivity of standalone firms' investment to both standalone and divisional q is significantly positive, with coefficients of 2.2% and 1.4%, respectively. The interaction terms of qs with C capture the incremental sensitivity of conglomerate divisions' investment. Consistent with Ozbas and Scharfstein (2010), we find a lower, albeit insignificantly so, sensitivity of divisional investment to standalone q of $2.2\% - 0.3\% = 1.9\%$. We find strikingly

¹⁰ Following the literature, if there are fewer than five standalone firms in a four-digit industry, we estimate the qs at the three-digit, or if needed the two-digit or one-digit, level.

Table VIII
The Investment- q Sensitivity of Standalone Firms and Conglomerate Divisions

This table investigates the sensitivity of investment to standalone and divisional qs . It reports results from panel regressions in which the dependent variable is the ratio of capital expenditures to sales of conglomerate divisions and stand-alone firms. The independent variables are the median q of standalone firms in the industry (Standalone q), calculated as in Berger and Ofek (1995), the divisional q in the Fama-French 10 industry (Divisional q), an indicator variable for conglomerate divisions (C), and their interactions. Columns (4) and (8) also control for the ratio of cash flows to sales. Columns (1) to (4) control for year and industry fixed effects, while columns (5) to (8) control for year and firm-division fixed effects. Standard errors are clustered by year, industry, and conglomerate status. t -Statistics are shown in brackets.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standalone q	0.022 [6.31]		0.023 [6.43]	0.022 [5.94]	0.015 [5.37]		0.015 [5.30]	0.015 [5.34]
Divisional q		0.014 [3.48]	0.003 [1.90]	0.004 [0.95]		0.010 [4.23]	0.004 [0.85]	0.004 [0.96]
C	0.010 [4.93]	−0.008 [−3.74]	−0.004 [−1.40]	−0.001 [−0.58]	0.001 [0.19]	−0.008 [−1.43]	−0.008 [−1.24]	−0.007 [−1.14]
Standalone $q \times C$	−0.003 [−1.27]		−0.009 [−2.03]	−0.007 [−1.49]	0.002 [0.60]		0.000 [−0.13]	0.000 [0.10]
Divisional $q \times C$		0.013 [4.11]	0.021 [3.87]	0.023 [4.82]		0.011 [2.40]	0.011 [6.70]	0.011 [6.91]
CFS				−0.041 [−2.67]				−0.058 [−3.61]
$CFS \times C$				−0.062 [−3.50]				−0.010 [−0.68]
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes				
Firm-division fixed effects					Yes	Yes	Yes	Yes
R^2	0.391	0.386	0.391	0.397	0.72	0.718	0.72	0.722
Number of observations	116,813	116,813	116,813	116,813	110,120	110,120	110,120	110,120

different results for divisional q , where conglomerates’ sensitivity increases significantly to $1.4\% + 1.3\% = 2.7\%$.

In regression (3), we include both measures of q , and in specification (4), we additionally control for the ratio of cash flows to sales.¹¹ Based on column (4), which includes year and industry fixed effects, the investment of standalone firms depends on standalone q but is insensitive to divisional q . In contrast, investment of conglomerate divisions is highly sensitive to divisional q and less sensitive to standalone q . In particular, the sensitivity of a standalone firm’s investment to standalone q is large and significant at 2.2%, while its exposure to divisional q is negligible (0.4%). In contrast, divisional investment is significantly more sensitive to our estimate of divisional q (2.3%, $t = 4.82$).

¹¹ Following Ozbas and Scharfstein (2010), we normalize cash flows by sales instead of assets because conglomerates may have more discretion in allocating assets across divisions.

We obtain similar results after including firm-division fixed effects. Based on column (8), the sensitivity of a standalone firm's investment to standalone q is large and significant at 1.5%, while its exposure to divisional q remains negligible (0.4%). In contrast, divisional investment remains more sensitive to our estimate of divisional q (1.1%, $t = 6.91$).¹²

These results indicate that compared to standalone firms, conglomerate investment is more sensitive to investment opportunities measured using conglomerate firms and less sensitive to investment opportunities estimated from standalone firms. This finding is consistent with the implications of Whited (2001), Maksimovic and Phillips (2002), and Gomes and Livdan (2004) that investment opportunities differ across conglomerates and standalone firms, and hence that using qs of standalone firms to proxy for divisional qs introduces measurement errors. These results suggest that prior findings should be interpreted with caution because they may arise due to measurement error in Tobin's q rather than inefficient allocation of capital inside conglomerates. The findings also imply that our method generates informative measures of conglomerate divisions' qs .

More broadly, the evidence suggests that by capturing the ability of conglomerates to mitigate external financing frictions through coinsurance offered by internal capital markets, divisional qs can better explain conglomerate investment. The analysis in Table VIII shows that the differences between standalone qs and divisional qs uncovered in the previous section have important implications for studying conglomerates' internal capital allocation. Unlike divisional qs , standalone qs do not account for coinsurance and covariances across conglomerate divisions, and hence cannot fully explain conglomerate investment.

B. Spinoffs

A possible concern is that the sensitivity of investment to divisional qs could be correlated with potentially unobservable firm- or division-level characteristics, and that these attributes, and not the organizational form, are responsible for the results in Table VIII. To address this concern, columns (5) to (8) already include firm-division fixed effects, which absorb time-invariant differences across divisions. Since these tests do not account for time-varying differences in divisional attributes, here we estimate the elasticity of investment to standalone and divisional qs around the spinoff of conglomerate divisions. This analysis delivers estimates that minimize such confounding effects as it traces

¹² In the Internet Appendix, we show that similar results obtain when conditioning on firms' cash holdings to account for theoretical predictions, suggesting that it is an important conditioning variable for conglomerate investment (e.g., Matvos and Seru (2014)). Since valuation multiples and equity returns depend on working capital and accruals, the quality of which might vary with organizational structure (Demirkan, Radhakrishnan, and Urcan (2012)), we also show robustness to estimating divisional qs after controlling for accruals, working capital, or noncash working capital.

Table IX
The Investment- q Sensitivity around Spinoffs

This table investigates the sensitivity of investment to standalone and divisional q s around the spinoffs of conglomerate divisions. It reports results from regressions in which the dependent variable is the ratio of capital expenditures to sales of the affected entities. The independent variables include the median q of standalone firms in the industry (Standalone q), calculated as in Berger and Ofek (1995), the divisional q in the Fama-French 10 industry (Divisional q), an indicator variable for the prespinoff period (PRE), and their interactions. For each spinoff, we include observations from three years before the spinoff to three years after the spinoff. Column (4) restricts the sample to spinoffs with at least one nonmissing investment observation in both the pre- and postspinoff periods, and column (5) requires at least two observations in each period. All regressions include year and industry fixed effects. Standard errors are clustered by industry and year. t -Statistics are shown in brackets.

	Firm Obs Required Both Pre- and Postspinoff				
	No Restriction			≥ 1	≥ 2
	(1)	(2)	(3)	(4)	(5)
Standalone q	0.040 [3.39]		0.044 [3.71]	0.049 [2.41]	0.052 [2.26]
Divisional q		0.014 [0.79]	-0.009 [-0.63]	0.005 [0.23]	0.017 [0.35]
PRE	0.023 [1.26]	-0.036 [-2.51]	-0.016 [-1.42]	-0.025 [-1.15]	-0.047 [-2.50]
Standalone $q \times PRE$	-0.012 [-1.47]		-0.026 [-3.13]	-0.027 [-1.80]	-0.028 [-1.70]
Divisional $q \times PRE$		0.032 [2.51]	0.051 [2.84]	0.062 [2.09]	0.080 [2.90]
R^2	0.137	0.113	0.141	0.148	0.203
Number of observations	1,551	1,551	1,551	1,057	711

the investment policy of the same entity over a relatively short time window surrounding the change in its organizational structure.

We focus on spun-off conglomerate divisions that start operating as public standalone firms. We collect these data by augmenting the spinoff data provided by John McConnell with a sample of spinoffs from the Securities Data Company (SDC).¹³ We then collect capital expenditures and sales data for six years around the event. We estimate the investment- q sensitivity regressions by introducing a prespinoff indicator (PRE) that allows the sensitivities to change at the time of the spinoff.

Table IX summarizes the results. In the first three columns, we do not impose any further data restrictions and we analyze 1,551 firm-year observations

¹³ McConnell's sample contains Center for Research in Security Prices (CRSP) permnos, which facilitate matching to Compustat. The SDC sample is matched to Compustat based on company name and manually verified. Compustat provides the information prior to spinoff for some firms, and we manually collect missing data from 10-K filings from Edgar when available. We thank John McConnell for making the data available on <http://www.krannert.purdue.edu/faculty/mcconnell/database>

that correspond to 452 spinoffs. The estimates in column (1) show that if divisional q is omitted, investment in the spun-off divisions is significantly related to standalone q after the spinoff with a sensitivity of 4.0%. Before the spinoff, when the firm was a division of a conglomerate, the sensitivity was 1.2% lower.

The evidence changes dramatically when we allow investment to depend on divisional qs . As shown in column (3), after the spinoff, investment of the newly formed standalone firm is significantly related to standalone q (4.4%, $t = 3.71$) and unrelated to divisional q . Prior to the spinoff, the then-division's investment is significantly less sensitive to standalone q at $4.4\% - 2.6\% = 1.8\%$, but significantly more sensitive to divisional q at $-0.9\% + 5.1\% = 4.2\%$. In other words, before the spinoff, divisional investment is highly related to qs estimated from conglomerates, whereas after the spinoff, it is only responsive to qs estimated from standalone firms.

In regressions (4) and (5), we repeat the analysis keeping only spinoffs with at least one or two nonmissing investment observations in both the pre- and postspinoff periods, respectively. Naturally, the sample size declines, but our results carry through and remain economically and statistically important. Indeed, the coefficient on the key interaction term of divisional q with PRE , which captures the importance of divisional qs for investment before the spinoff, increases from 5.1% in column (3) to 6.2% and 8.0% in columns (4) and (5), respectively.

Overall, these results suggest that a firm's investment opportunities, measured by Tobin's q , change when its organizational structure changes from a conglomerate division to a standalone firm. Consequently, its investment policy, as measured by its investment- q elasticity, changes as well. As this analysis holds other firm attributes constant, it emphasizes the crucial role of organizational structure in estimating qs .

C. Diversifying Acquisitions

The results thus far show that divisional qs , which capture the ability of conglomerates to mitigate the effects of credit shocks, explain the variation in conglomerates' internal capital allocation. We now investigate reallocation of assets across firms via mergers and acquisitions. We conjecture that the time-varying differences between divisional and standalone qs capture the variation in the value of conglomeration relative to focused organizational structures amidst varying credit conditions and external financing frictions, and consequently explain merger and acquisition activity as well as acquisition announcement returns. In particular, RDQs in a given industry increase when operating as part of a conglomerate is more appealing. Consequently, RDQs should positively relate to volume and announcement returns of diversifying acquisitions.

We study acquisition activity in Table X. Every year, for each of the 10 Fama-French industries, we count the number of diversifying acquisitions in which acquirer, target, or either acquirer or target belongs to the industry. Our 1978 to 2014 sample comes from SDC and includes all mergers between

Table X
Acquisition Activity

This table reports results of panel regressions of the logarithm of one plus the number of diversifying acquisitions in an industry-year on the median q of standalone firms and the relative divisional q (RDQ) in the Fama-French 10 industry. The sample requires both target and acquirer to be public standalone firms, and the acquisition to be diversifying across two three-digit SIC industries. All regressions include industry fixed effects. Standard errors are clustered by industry. t -Statistics are shown in brackets.

	Standalone q	RDQ	R^2
Number of acquirers by industry	0.329 [3.65]	0.437 [3.39]	0.572
Number of acquisition targets by industry	0.308 [2.81]	0.327 [2.27]	0.629
Number of targets or acquirers by industry	0.316 [2.64]	0.380 [2.76]	0.570

public standalone firms in distinct three-digit SIC industries. We estimate panel regressions in which the dependent variable is the logarithm of one plus the number of transactions in an industry-year. The key independent variable, RDQ, is lagged by one year to mitigate concerns about the simultaneity of q s and acquisition activity. All regressions include industry fixed effects.

The significantly positive and economically meaningful coefficients on RDQ suggest that higher values of conglomeration relative to focused organizational structures correspond to greater diversifying acquisition activity in the next year. For example, when using the total number of acquirers and targets in an industry, the coefficient on RDQ is 0.380 ($t = 2.76$). This estimate implies that an increase of 0.22 in RDQ, its average standard deviation from Table II, corresponds to an 8.5% increase in the number of diversifying acquisitions. These results highlight the crucial role that systematic differences between q s of conglomerate divisions and standalone firms play in real asset allocation in the economy.

In the final set of analyses, Table XI reports results on acquisition announcement returns. In regressions (1), (2), (4), (5), and (7), we consider mergers that are diversifying across three-digit SIC industries as above. In specifications (3), (6), and (8), we study a smaller sample of mergers that diversify across the 10 Fama-French industries. The dependent variables are the cumulative stock returns in excess of market returns of acquirers (columns (1) to (3)), targets ((4) to (6)), or the combined firm ((7) and (8)) over the five-day window centered on the announcement day. The independent variable of interest is again the RDQ in the previous year, and all regressions control for log market capitalization, profitability, and asset growth of both targets and acquirers and include year fixed effects. Returns and RDQs for the combined firm are market capitalization-weighted averages.

Table XI

Acquisition Announcement Returns

This table analyzes the relation between q s and acquisition announcement returns. Each column reports estimates from a regression in which the dependent variable is the acquisition announcement return of acquirers, targets, or the combined firm (i.e., the value-weighted average). Returns are cumulated over the five-day period centered around the announcement date and are in excess of market returns. The independent variables include the median q of standalone firms in the industry, calculated as in Berger and Ofek (1995), as well as the relative divisional q (RDQ). All regressions also control for log market capitalization, profitability, and asset growth of both target and acquirer. All independent variables are measured in the year prior to the announcement year. The sample requires both target and acquirer to be public standalone firms, and the acquisition to be diversifying across 2 three-digit SIC industries (columns (1), (2), (4), (5), and (7)) or across two Fama-French 10 industries ((3), (6), (8)). All regressions include year fixed effects and cluster standard errors by year. t -Statistics are shown in brackets.

	Acquirer Return			Target Return			Combined Return	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standalone q of the acquirer	0.021 [3.93]							
Standalone q of the target				−0.006 [−0.26]				
Standalone q of the combined firm		0.019 [2.71]	0.025 [2.68]		0.043 [2.70]	0.070 [3.14]	0.016 [2.74]	0.027 [3.85]
RDQ of the acquirer	0.056 [3.69]							
RDQ of the target				−0.056 [−1.42]				
RDQ of the combined firm		0.050 [2.62]	0.055 [3.46]		0.064 [1.61]	0.100 [1.77]	0.045 [2.39]	0.056 [4.22]
SIC three-digit diversification	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fama-French 10 diversification		0.174 336	0.297 143	0.199 336	0.201 336	0.307 143	0.175 336	0.322 143
R^2	0.204 336							
Number of observations								

The main takeaway from Table XI is that RDQs positively predict announcement returns of diversifying acquisitions. This result holds for acquirers, targets, and the combined firm, and is both statistically significant and economically important. For example, based on specification (7), a one-standard-deviation increase in the RDQ of the combined firm corresponds to an increase of 0.8% in the announcement return of the combined firm.

Interpreted broadly, the results in this subsection suggest that the differences between divisional and standalone qs , which capture time-varying benefits of conglomerates' internal capital markets amidst changing credit conditions, predict both the intensity of, and the market's reaction to, diversifying acquisitions. As such, they suggest that the differences between divisional and standalone qs are important determinants of cross-firm asset allocation via the market for corporate control.

V. Conclusion

We propose a new method to estimate the valuation multiples, or qs , of conglomerate divisions. Our approach uses only standard data and does not rely on standalone firms, thus mitigating concerns about measurement errors that may arise due to unobservable differences between conglomerate divisions and standalone firms. The resulting estimates of intraconglomerate qs allow us to study the systematic differences between conglomerates and standalone firms and their implications for asset allocation within and across firms.

We find that qs of conglomerate divisions differ considerably from commonly used industry-level standalone qs . Divisional qs vary over time and across industries, and divisional valuations range from deep discounts to large premiums relative to those of standalone firms. We investigate the economic mechanisms that explain these differences and find that divisional qs are less sensitive to economic shocks compared to qs of standalone firms. We show that this reduced sensitivity arises because coinsurance provided by internal capital markets of conglomerates mitigates the effects of external financing frictions and credit shocks.

In the final set of analyses, we show that differences in qs affect the allocation of resources inside firms via internal capital markets and across firms via diversifying acquisitions. First, we show that conglomerate investment and internal capital allocation are highly sensitive to divisional qs , and this sensitivity changes following spinoffs of conglomerate divisions. Second, we find that the difference between divisional and standalone qs , which captures time-varying benefits of conglomeration, predicts the volume and announcement returns of diversifying acquisitions.

Overall, our method provides disaggregated estimates of division-level valuation multiples, which shed new light on the attributes and policies of conglomerates. Given that our approach estimates valuation multiples at the granular level of a conglomerate division, it can be extended to broaden our understanding of corporate diversification along additional economic channels, including cost structures, supplier networks, bargaining power, competitive-

ness, innovation, and synergistic spillovers across product markets and supply chains. For example, future research can use our method, combined with establishment-level data on assets, productivity, and employees, to generate disaggregated qs at the establishment level to study valuation and internal resource allocation at more granular levels inside conglomerates. More generally, our approach can be adapted to other applications where unobservable attributes of parts may be extracted from their observable analog of the whole.

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Appendix A: Industry Definitions

The 10 Fama-French industries are defined as follows:

1. NDur: Consumer nondurables—food, tobacco, textiles, apparel, leather, toys.
2. Dur: Consumer durables—cars, TVs, furniture, household appliances.
3. Manu: Manufacturing—Machinery, trucks, planes, chemicals, office furniture, paper, commercial printing.
4. Enrg: Oil, gas, and coal extraction and products.
5. HiTec: Business equipment—computers, software, and electronic equipment.
6. Tlcm: Telephone and television transmission.
7. Shop: Wholesale, retail, and some services (laundries, repair shops).
8. Hlth: Healthcare, medical equipment, and drugs.
9. Util: Utilities.
10. Oth: Other—mines, construction, building materials, transportation, hotels, business services, entertainment, finance (except SIC 6000-6999, 8600, 8800, 8900, 9000).

Appendix B: Variable Definitions

- Assets: Total assets (Compustat data item AT for firms, IAS for divisions).
- CDS spread: the annualized premium needed to compensate the counterparty for default risk, on an actuarial basis, over a one-year horizon. The industry CDS is the median of December values across all single-segment firms in the industry. Data are provided by the Credit Research Initiative (CRI) database of the National University of Singapore, available at <http://rmicri.org>; accessed July 5, 2020 and available starting in 1990.
- CFS: the cashflow to sales ratio is the sum of operating profits and depreciation (Compustat “OPS,” “DPS”), divided by total sales.
- Coinsurance potential: to estimate coinsurance potential, we first calculate the full-sample time-series covariance of annual sales growth of standalone firms for each three-digit SIC industry pair. Matching these estimates to divisions, we construct the sales-weighted covariance matrix

for each conglomerate-year. We then calculate the relative contribution of a single division to the conglomerate's variance by dividing the sum of all matrix elements that are related to this division by the sum of all elements. High coinsurance divisions are defined as those with below-median relative contribution to conglomerate variance.

- Credit rating: the last credit rating within the fiscal year provided by Standard & Poor's. Ratings are coded to numerical values so that low values correspond to low credit risk ($AAA = 1, AA+ = 2, \dots D = 22$). The industry credit rating is the mean across all single-segment firms in the industry. Data are obtained from Compustat ("splticrm") and available from 1985 to 2017.
- Credit spread: firm-level corporate bond yields are the average yields of bonds on the last day they traded in the calendar year. Bonds that do not trade between September and December are discarded. Yields are in excess of the 10-year Treasury constant-maturity yield from Federal Reserve Economic Data (FRED, "GS10"). The industry credit spread is the median across all single-segment firms in the industry. Data are obtained from TRACE and available starting in 2004.
- Default spread: Moody's seasoned Baa corporate bond yields minus the 10-year Treasury constant-maturity yield, averaged within calendar year. Data are from FRED ("BAA," "GS10").
- Distance to default: we compute the firm-level Merton (1974) distance to default as proposed by KMV and implemented using the algorithm of Bharath and Shumway (2008). The industry distance to default is the median of December values across all single-segment firms in the industry.
- Dividend yield: for each month, we compute the sum of dollar dividends paid by all stocks in the S&P 500 over the past 12 months divided by the index level, and average within calendar year.
- Expansion indicator: an indicator variable equal to one if the month is an NBER expansion and zero otherwise and averaged within calendar year. Data are from FRED ("USREC").
- Firm value: Book value of debt (Compustat DLTT + DLC) plus market value of equity (Compustat CSHO $\times$ PRCC_F).
- Industrial production: an economic indicator that measures production of manufacturing, mining, and electric and gas utilities facilities located in the U.S. Data are from FRED ("INDPRO").
- Industry dividend yield: for each month, we compute the sum of dollar dividends paid by all single-segment firms in the industry over the past 12 months divided by the sum of their market values of equity, and average within calendar year.
- Industry return: value-weighted calendar-year return of single-segment firms in the industry.
- Investment: the investment rate is computed as capital expenditures (Compustat CAPX, CAPXS) divided by total sales.
- Market return: calendar-year return of the CRSP value-weighted market index.

- Output: Δ Output is the aggregate output shock from John Fernald's website (<http://www.johnfernald.net/TFP>). The quarterly values are averaged within calendar year.
- Sales: Total sales (Compustat SALE, SALES).
- TED spread: average daily TED spread level within calendar year. Data are from FRED ("TEDRATE," from 1986 onward).
- TFP: Δ TFP is the business sector total factor productivity shock from John Fernald's website (<http://www.johnfernald.net/TFP>). The quarterly values are averaged within calendar year.
- VIX: month-end VIX levels are averaged within calendar year. Data are from CBOE (1986 to 1989) and FRED ("VIXCLS," from 1990 onward).

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.